STUDENT PERFORMANCE ANALYSIS REPORT

Subject: Data Fundamentals with Data Warehousing

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1. Problem Statement

In educational institutions, managing and analyzing student performance data is often a time-consuming and inefficient process when done manually. Teachers and administrators require a systematic approach to evaluate student performance, identify top performers, and detect students who need improvement.

Traditional methods lack:

- Efficient data organization
- Quick analysis capabilities
- Visual insights for better decision-making

Hence, there is a need for a system that can efficiently store, process, and analyze student data to generate meaningful insights.

2. Proposed Solution

To address this problem, a **Student Performance Analysis System** has been developed using SQL, Excel, and Python. This system integrates data storage, processing, and visualization techniques to provide a comprehensive analysis of student performance.

Key Features of the Solution:

- Centralized data storage using SQL
- Automated calculation of average marks
- Identification of top-performing students
- Detection of students scoring below a threshold
- Department-wise performance comparison
- Visualization of data using charts and graphs
- Correlation analysis between attendance and performance

The system ensures efficient data handling and enables quick decision-making.

3. Technologies Used (Tech Stack)

The project uses a combination of tools and technologies:

1. SQL (MySQL Workbench)

- Used for database creation and management
- Performs structured queries for analysis

2. Microsoft Excel

- Used for data visualization and formatting
- Supports pivot tables and conditional formatting

3. Python

- Libraries used:
    - Pandas (data manipulation)
    - Matplotlib (visualization)

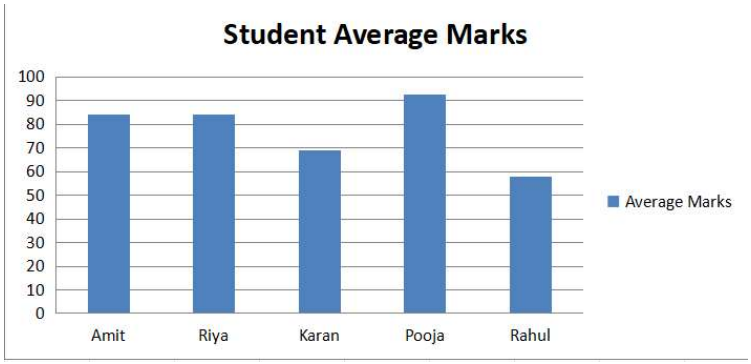
4. CSV File

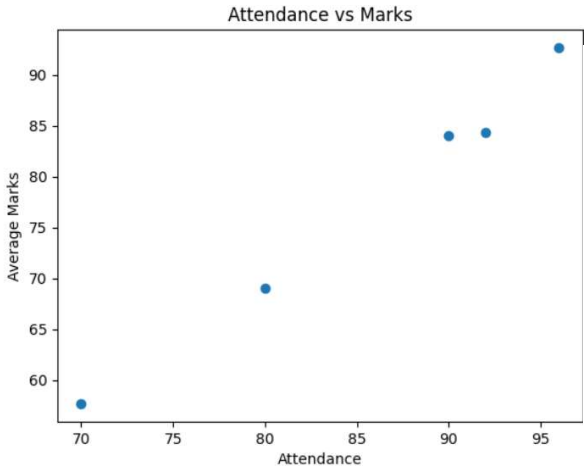
- Used as the primary dataset format for data exchange

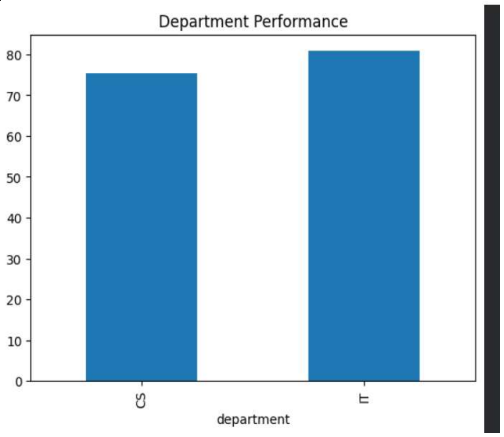
4. Screenshots

Student Performance Analysis

student_id	name	department	math	science	programming	attendance	Average Marks
1	Amit	CS	78	85	90	92	84.33
2	Riya	CS	88	79	85	90	84
3	Karan	IT	65	70	72	80	69
4	Pooja	IT	92	91	95	96	92.67
5	Rahul	CS	55	60	58	70	57.67








```
••• Top Performer:
     student_id               4
    name                  Pooja
    department               IT
    math                     92
    science                  91
    programming              95
    attendance               96
    average_marks    92.666667
    Name: 3, dtype: object

    Department Average:
     department
    CS    75.333333
    IT    80.833333
    Name: average_marks, dtype: float64

    Students at Risk:
        student_id   name department  math  science  programming  attendance  \
    4            5  Rahul         CS    55       60           58          70

       average_marks
    4      57.666667

    Correlation: 0.9955648268660175
```

4. Unique Points of the Project

This project stands out due to the following features:

- Integration of multiple tools (SQL + Excel + Python)
- End-to-end data analysis workflow
- Combination of structured queries and visual insights
- Identification of both high-performing and at-risk students
- Correlation analysis for deeper understanding
- Simple yet effective design suitable for real-world use

5. Future Improvements

The project can be enhanced further by:

- Expanding dataset with more students and subjects
- Adding a user interface (web or app-based system)
- Implementing real-time data updates
- Using machine learning for performance prediction
- Adding dashboards using tools like Power BI or Tableau
- Integrating cloud-based databases

6. Conclusion

The Student Performance Analysis System successfully demonstrates how data can be transformed into meaningful insights using modern tools. By combining SQL, Excel, and Python, the project provides a complete pipeline for data storage, processing, analysis, and visualization.

This system can be further extended to support larger datasets and more advanced analytics, making it a valuable tool for academic institutions.